

BARKSDALE AFB, LA, USA

WMO: 722485

Lat: 32.500N

Lon: 93.667W

Elev: 166

StdP: 14.61

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 13944

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	24.2	27.5	10.4	9.4	30.3	15.2	11.9	34.4	24.5	54.4	22.0	52.2	6.1	330	0.425

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.1	99.1	75.5	96.6	75.7	94.4	75.9	79.3	90.3	78.4	89.1	77.9	88.4	7.0	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.1	141.9	81.2	76.1	136.9	81.3	75.2	133.1	81.3	42.7	90.3	42.0	88.9	41.3	88.6	90.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
20.5	18.3	16.3		DB	17.9	101.2	4.2	3.9	14.9	104.0	12.4	106.3	10.1	108.4	7.0	111.3
				WB	16.4	81.1	4.1	2.4	13.4	82.8	11.0	84.2	8.7	85.5	5.7	87.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	65.8	46.8	51.3	58.0	64.9	73.2	80.3	83.1	83.0	77.2	66.3	55.7	49.4
	DBStd	15.05	10.32	10.28	9.27	7.57	6.05	4.33	3.92	4.46	5.76	7.86	9.04	9.86
	HDD50	510	186	100	36	3	0	0	0	0	0	2	45	138
	HDD65	2198	569	394	252	94	11	0	0	0	3	82	300	493
	CDD50	6290	87	136	283	450	720	910	1025	1022	816	508	215	118
	CDD65	2504	4	10	34	92	266	460	560	557	370	123	19	9
	CDH74	26174	13	56	249	740	2369	4893	6529	6435	3627	1105	125	33
	CDH80	12204	0	3	34	156	855	2321	3357	3405	1704	351	18	0
Wind	WSAvg	7.1	8.1	8.3	8.6	8.9	7.7	6.7	6.0	5.6	5.4	6.0	6.9	7.5
Precipitation	PrecAvg	51.7	4.9	4.2	5.3	5.0	4.4	4.3	3.6	3.0	3.7	4.0	4.3	4.9
	PrecMax	74.6	13.2	8.3	14.8	16.8	10.1	13.1	10.5	8.4	8.0	17.4	10.7	8.1
	PrecMin	28.9	0.4	0.2	0.9	1.2	0.6	0.4	0.6	0.2	0.3	0.0	1.0	0.4
	PrecStd	11.5	2.8	2.3	3.0	3.5	2.8	3.0	2.4	2.1	2.2	3.6	2.4	1.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	75.5	79.6	83.6	88.1	93.1	99.3	100.5	103.7	99.0	90.7	82.4	77.5
		MCWB	63.7	64.4	67.1	70.0	73.8	75.4	75.2	75.2	73.3	71.7	69.5	67.3
	2%	DB	72.2	75.3	80.2	84.1	90.1	96.0	98.2	99.7	95.3	87.2	77.4	73.6
		MCWB	61.2	63.6	65.7	69.4	73.4	75.6	75.6	76.1	74.0	70.8	66.3	65.4
	5%	DB	68.1	71.7	76.6	81.5	87.7	93.2	95.5	96.7	92.2	84.2	73.7	70.0
		MCWB	59.9	62.7	63.8	67.4	73.1	75.9	76.4	76.2	74.1	68.9	64.6	64.4
	10%	DB	63.8	68.0	72.9	78.7	85.3	90.8	93.2	94.1	89.6	80.7	70.5	66.3
		MCWB	57.9	60.5	62.3	66.3	72.3	75.9	76.6	76.1	73.5	68.1	63.7	61.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	66.8	68.8	70.1	74.1	77.7	79.6	80.1	80.7	78.8	76.3	71.9	70.1
		MCDB	72.5	73.7	77.3	81.3	87.1	88.7	91.0	92.6	87.7	84.3	77.7	73.6
	2%	WB	64.5	66.6	68.3	72.1	76.1	78.4	79.0	79.4	77.5	74.8	69.6	67.9
		MCDB	69.4	71.9	75.5	79.5	85.2	88.0	90.5	90.9	86.9	82.2	74.5	71.6
	5%	WB	62.4	64.8	66.7	70.7	74.8	77.7	78.3	78.5	76.6	73.1	67.6	65.9
		MCDB	66.3	69.7	73.5	77.8	83.7	87.7	89.4	89.4	85.7	79.2	72.0	69.5
	10%	WB	59.2	62.4	64.9	69.2	73.7	77.0	77.6	77.9	75.4	71.0	64.7	62.7
		MCDB	62.4	66.7	71.3	75.9	82.3	86.8	88.5	88.5	84.3	76.5	69.2	65.8
Mean Daily Temperature Range	5% DB	MDBR	20.3	20.4	21.2	21.8	20.0	19.1	19.1	20.2	21.7	23.0	21.5	19.8
		MCDBR	25.5	24.0	23.9	23.5	21.7	21.6	21.4	23.6	24.4	25.8	24.1	23.1
		MCWBR	16.7	14.8	12.0	10.8	7.8	5.9	5.2	6.2	7.3	10.8	14.5	15.9
	5% WB	MCDBR	19.7	19.9	20.1	18.9	18.6	18.5	18.9	19.7	19.6	19.7	20.5	20.3
MCWBR		15.4	14.5	12.3	10.8	7.9	6.6	6.2	6.4	6.7	10.5	14.7	16.1	
Clear-Sky Solar Irradiance	taub	0.331	0.342	0.366	0.399	0.429	0.463	0.477	0.476	0.422	0.365	0.351	0.335	
	taud	2.502	2.470	2.410	2.318	2.294	2.240	2.230	2.226	2.346	2.479	2.467	2.494	
	Ebn at Noon	282	290	289	283	274	263	259	258	268	278	272	274	
	Edh at Noon	28	31	36	41	42	44	45	44	38	31	28	26	
All-Sky Solar Radiation	RadAvg	855	962	1361	1694	1871	1983	1983	1862	1597	1289	935	743	
	RadStd	84	162	119	108	134	148	148	159	143	177	103	96	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	+1.37	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (31 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page